FINAL PROJECT REPORT

Heart Disease Analysis using Tableau Dashboard

1. INTRODUCTION

1.1 Project Overview

Heart disease is one of the leading causes of death worldwide. Identifying risk factors and understanding patterns in heart disease data is essential for improving healthcare outcomes. With the increasing availability of healthcare data, data visualization tools can help extract meaningful insights from complex datasets.

This project focuses on analyzing heart disease data using **Tableau**, a powerful data visualization tool that allows users to create interactive dashboards and graphical representations of data. The dataset contains health-related indicators such as **Body Mass Index (BMI), smoking habits, alcohol consumption, age category, sleep duration, diabetes, physical activity levels, and other medical conditions** that may influence heart health.

The main objective of this project is to explore the relationships between these lifestyle and health factors and the occurrence of heart disease. By organizing and visualizing the dataset through charts, graphs, and filters, the project makes it easier to identify patterns that might not be visible in raw data tables.

Data visualization plays an important role in modern healthcare analytics. Instead of manually examining large volumes of data, visual dashboards help users quickly understand trends, correlations, and distributions. Tableau provides features such as **interactive filters, dynamic charts, and real-time data exploration**, which make it suitable for analyzing healthcare datasets.

By using interactive visualizations, the project aims to identify patterns and correlations between lifestyle factors and heart disease risk. For example, the dashboard can show how heart disease cases vary across different age groups, how smoking habits affect heart disease risk, or how BMI and physical activity levels are related to cardiovascular health.

The dashboard provides healthcare professionals and researchers with an easy-to-understand interface to analyze the data and make informed decisions. It allows users to explore different

aspects of the dataset through filters and visual comparisons, making the analysis process faster and more efficient.

Overall, this project demonstrates how **data visualization techniques can transform complex healthcare data into meaningful insights**. By presenting the information in a clear and interactive format, the dashboard helps users better understand the factors that contribute to heart disease and supports data-driven decision-making in healthcare research and prevention strategies.

1.2 Purpose

The main purpose of this project is to analyze heart disease risk factors and present insights using an interactive Tableau dashboard. The project helps in understanding how various lifestyle habits and health conditions influence the likelihood of heart disease.

Heart disease is influenced by many factors such as age, lifestyle choices, medical history, and daily habits. By analyzing healthcare data, it becomes possible to identify patterns and trends that indicate a higher risk of heart disease. This project uses data visualization techniques to transform complex datasets into clear and meaningful graphical representations.

The Tableau dashboard allows users to explore the dataset interactively by applying filters and viewing different charts. This makes it easier to understand relationships between different health indicators and heart disease occurrence. The insights generated through this analysis can support researchers, healthcare professionals, and policy makers in identifying high-risk groups and planning preventive measures.

In addition, the project demonstrates how visualization tools can simplify large datasets and make data analysis more accessible to users who may not have advanced statistical knowledge. By presenting information in an interactive and visual format, the dashboard helps users quickly identify important trends and correlations.

The objectives of this project include:

- **Analyzing heart disease patterns using healthcare data.**
- **Identifying key risk factors such as smoking, BMI, and physical activity.**
- **Creating interactive visualizations using Tableau.**
- **Helping healthcare professionals interpret health data effectively.**
- **Supporting data-driven decision-making for disease prevention.**
- **Understanding the relationship between lifestyle habits and cardiovascular health.**

- **Providing an easy-to-use visual platform for exploring healthcare data.**

Overall, the purpose of this project is to demonstrate how data visualization can help simplify complex medical data and support better understanding of the factors that contribute to heart disease.

2. IDEATION PHASE

2.1 Problem Statement

PS-1

I am a healthcare analyst trying to identify key risk factors contributing to heart disease, but the dataset contains many variables that are difficult to interpret manually. Because the relationships between lifestyle habits and heart disease are not clearly visible in raw data, it becomes challenging to draw meaningful insights, which makes me feel frustrated while analyzing healthcare information.

In addition, healthcare datasets often include multiple attributes such as age, gender, BMI, smoking habits, alcohol consumption, physical activity levels, and medical conditions. Analyzing these variables individually does not provide a complete understanding of how they interact with each other. Without proper visualization tools, identifying patterns and correlations between these factors becomes time-consuming and inefficient.

Therefore, there is a need for a system that can simplify the analysis process and present the data in a visual format. Interactive dashboards can help analysts quickly observe trends, compare different variables, and identify the most significant risk factors associated with heart disease.

PS-2

I am a public health researcher trying to understand how lifestyle habits such as smoking, alcohol consumption, sleep time, and physical activity influence heart disease risk. However, traditional data analysis methods do not clearly highlight patterns because the data is complex and contains multiple attributes, which makes me feel the need for better visualization tools.

Public health researchers often rely on data-driven insights to design prevention strategies and health awareness programs. However, when the data is presented only in tabular format, it becomes difficult to quickly identify which factors contribute most to heart disease. This limitation reduces the efficiency of research and decision-making.

By using advanced visualization tools such as Tableau, complex healthcare data can be transformed into interactive charts and dashboards. These visualizations help researchers explore the data more effectively, compare different factors, and gain a clearer understanding of heart disease trends across different groups of people.

2.2 Empathy Map Canvas

User: Healthcare analysts and medical researchers.

Thinks

- Which factors increase heart disease risk?
- Is smoking strongly related to heart disease?
- Which age group is most affected?
- How do lifestyle habits influence cardiovascular health?

Feels

- Concerned about increasing heart disease cases.
- Needs accurate insights for decision making.
- Wants a clear understanding of health data.
- Motivated to find patterns that can help prevent diseases.

Says

- Visualization tools help understand data faster.
- Lifestyle habits should be analyzed properly.
- Healthcare data should be presented in a simple and clear format.

Does

- Analyzes healthcare datasets.
- Uses visualization tools like Tableau.
- Studies patterns between health indicators and diseases.
- Creates reports and dashboards for data analysis.

Pains

- Large datasets are difficult to interpret.
- Raw data does not easily show patterns.
- Traditional analysis methods are time-consuming.

Gains

- Interactive dashboards for analysis.
- Clear visualization of risk factors.
- Faster understanding of health trends.
- Better insights for healthcare decision-making.

2.3 Brainstorming

During the brainstorming session, the team discussed various ideas for analyzing heart disease data and visualizing the results effectively. The main goal was to identify meaningful insights from the dataset and present them in a way that is easy to understand for users.

The team explored different possible visualization techniques such as bar charts, pie charts, line charts, and interactive dashboards. Various health indicators in the dataset were considered to determine how they might influence heart disease risk.

Generated Ideas

- Analyze heart disease cases by age category.
- Compare smokers vs non-smokers.
- Study the relationship between BMI and heart disease.
- Analyze the impact of physical activity.
- Study the effect of sleep duration.
- Compare heart disease cases across genders.
- Analyze diabetes and heart disease relationship.
- Study the influence of general health conditions on heart disease.

Selected Ideas

The team prioritized ideas based on feasibility, relevance, and the potential impact on understanding heart disease risk factors.

Selected visualizations for the dashboard include:

- **Heart Disease by Age Category**
- **Heart Disease by Gender**
- **Heart Disease by Smoking Habit**
- **Heart Disease by BMI**
- **Heart Disease by Physical Activity**
- **Heart Disease by Sleep Duration**

These visualizations help provide a comprehensive overview of the factors associated with heart disease and allow users to explore the data interactively through the Tableau dashboard.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

User opens the Tableau dashboard.
User selects filters such as age category or gender.
Dashboard displays visualizations related to heart disease.
User analyzes patterns between health indicators and heart disease.
User interprets insights to understand risk factors.

The customer journey begins when the user accesses the Tableau dashboard designed for heart disease analysis. The dashboard provides an intuitive interface where users can interact with different visual elements such as charts, graphs, and filters.

Users can apply filters based on attributes such as **age category, gender, smoking habits, BMI levels, or physical activity**. Once the filters are applied, the dashboard dynamically updates the visualizations to display relevant data.

By interacting with the dashboard, users can explore relationships between different health indicators and the occurrence of heart disease. This interactive exploration allows healthcare analysts and researchers to easily detect patterns, trends, and correlations within the dataset.

Finally, the user interprets the insights obtained from the visualizations to identify possible risk factors and draw meaningful conclusions that can support healthcare research and decision-making.

3.2 Solution Requirement

Functional Requirements

Import dataset into Tableau.
Create charts and graphs for analysis.
Implement filters for age, gender, and lifestyle factors.
Display heart disease statistics visually.

Additional functional capabilities include allowing users to interact with the dashboard through dynamic filtering and visualization updates. The system should support multiple chart types such as bar charts, pie charts, and comparison charts to represent different aspects of the dataset.

The dashboard should also allow users to quickly compare different variables and identify relationships between health indicators and heart disease risk.

Non-Functional Requirements

Easy-to-understand dashboard.
Fast visualization rendering.
Interactive filters for better exploration.

The dashboard should be designed in a user-friendly manner so that even users without technical expertise can easily navigate and understand the visualizations. Performance is also important, so the dashboard should load quickly and respond smoothly when filters are applied.

Additionally, the visualizations should be clear, well-organized, and visually appealing to ensure that users can interpret the information efficiently.

3.3 Data Flow Diagram

Level 0 (Overview)

Dataset → Data Processing → Tableau Visualization → Dashboard Insights

This high-level diagram represents the overall flow of data within the system. The dataset is first collected and processed before being visualized in Tableau. The final output is presented as insights through an interactive dashboard.

Level 1

CSV Dataset
↓
Data Cleaning and Processing
↓
Data Visualization using Tableau
↓
Dashboard Creation
↓
User Interaction and Analysis

At this level, the diagram provides a more detailed representation of the process. The raw dataset is initially stored in CSV format. The data is then cleaned and prepared to ensure accuracy and consistency.

After preprocessing, the data is imported into Tableau where different visualizations are created. These visualizations are then combined to form an interactive dashboard. Finally, users interact with the dashboard to analyze the data and gain insights about heart disease risk factors.

3.4 Technology Stack

Technology	Purpose
Tableau	Data visualization and dashboard creation
CSV Dataset	Data source
Excel	Data inspection and preprocessing
Windows OS	Development environment

The technology stack used in this project focuses mainly on data visualization and data preparation. Tableau serves as the primary tool for building interactive dashboards and presenting insights.

The dataset is stored in CSV format, which makes it easy to import into various tools. Excel is used for initial data inspection, cleaning, and formatting before importing the data into Tableau. The project is developed and executed in a Windows operating system environment.

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4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem of analyzing complex healthcare data can be solved using visualization techniques. Healthcare datasets often contain large amounts of information that are difficult to interpret using traditional methods.

Tableau provides interactive dashboards that help users understand patterns in data quickly. By converting raw data into visual formats such as charts and graphs, users can easily identify trends and relationships between different variables.

By applying visual analytics to heart disease data, the project helps identify important risk factors and provides a clearer understanding of how lifestyle habits and medical conditions influence heart disease risk. This approach simplifies data interpretation and supports effective healthcare analysis.

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard that visualizes heart disease data using different charts and filters.

The dashboard will allow users to:

Analyze heart disease distribution across age groups.
Compare heart disease cases between genders.
Study the relationship between smoking and heart disease.
Analyze how BMI and physical activity affect heart disease risk.

In addition, the dashboard will allow users to interactively explore the dataset by applying filters and selecting different variables. This flexibility helps users focus on specific aspects of the data and gain deeper insights.

The visualizations are designed to present complex information in a simple and intuitive way so that users can quickly understand the relationships between different health indicators.

4.3 Solution Architecture

Architecture Flow

Dataset (CSV File)
↓
Data Cleaning and Preparation
↓
Data Import into Tableau
↓
Creation of Visualizations
↓
Dashboard Integration
↓
User Interaction and Analysis

The solution architecture describes how the data flows through different stages of the project. Initially, the dataset is collected and stored in CSV format. The data is then cleaned and prepared to remove inconsistencies or missing values.

After preprocessing, the cleaned dataset is imported into Tableau where different visualizations are created. These visualizations are then integrated into a single interactive dashboard.

Finally, users interact with the dashboard to explore the data, analyze patterns, and gain insights into heart disease risk factors.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Description
Dataset Collection	Obtain the heart disease dataset from a reliable source.
Data Understanding	Analyze dataset attributes such as BMI, smoking habits, age category, sleep time, and other health indicators.
Visualization Design	Decide the types of charts and dashboard layout for effective data representation.
Dashboard Development	Build the dashboard using Tableau and create interactive visualizations.
Testing	Validate dashboard outputs and ensure charts and filters work correctly.

Final Report	Prepare project documentation including analysis, results, and conclusions.

The project planning phase was designed to ensure that the development process was structured and organized. Each phase focused on a specific task, starting from collecting the dataset to preparing the final report.

During the planning stage, the team identified the key variables in the dataset and determined the most suitable visualization techniques to represent the data. This structured approach helped ensure that the project was completed efficiently and met the objectives of analyzing heart disease risk factors.

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard was tested to ensure proper functionality and performance. Testing was conducted to verify that all visualizations, filters, and dashboard components worked correctly.

Testing Criteria

Test Case	Result
Dataset Loading	Successful
Chart Generation	Successful

Filters Working	Successful
Dashboard Interaction	Successful

The testing process confirmed that the dashboard loads the dataset correctly and generates the required charts without errors. The interactive filters function properly, allowing users to explore different aspects of the data.

Additionally, the dashboard was evaluated for performance to ensure that it responds quickly when filters are applied or when users interact with different visual elements. The results show that the dashboard performs efficiently and provides quick visual insights.

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7. RESULTS

The project successfully created a **Tableau dashboard** that analyzes heart disease data and presents insights visually. The dashboard allows users to interact with the dataset and explore relationships between different health indicators and heart disease risk.

By using interactive charts and filters, users can easily analyze how lifestyle factors such as smoking, physical activity, and BMI influence heart disease.

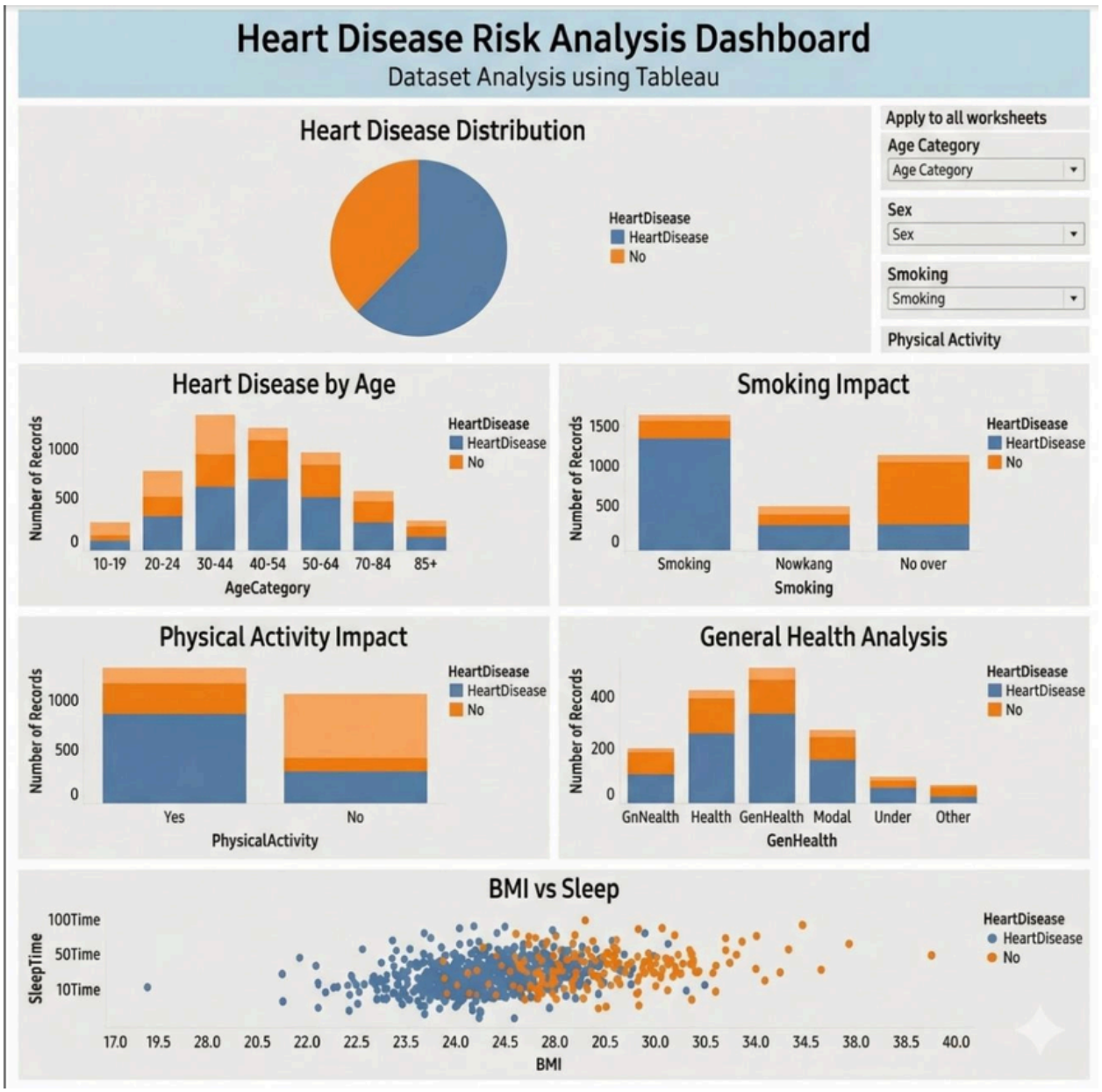
Key Observations

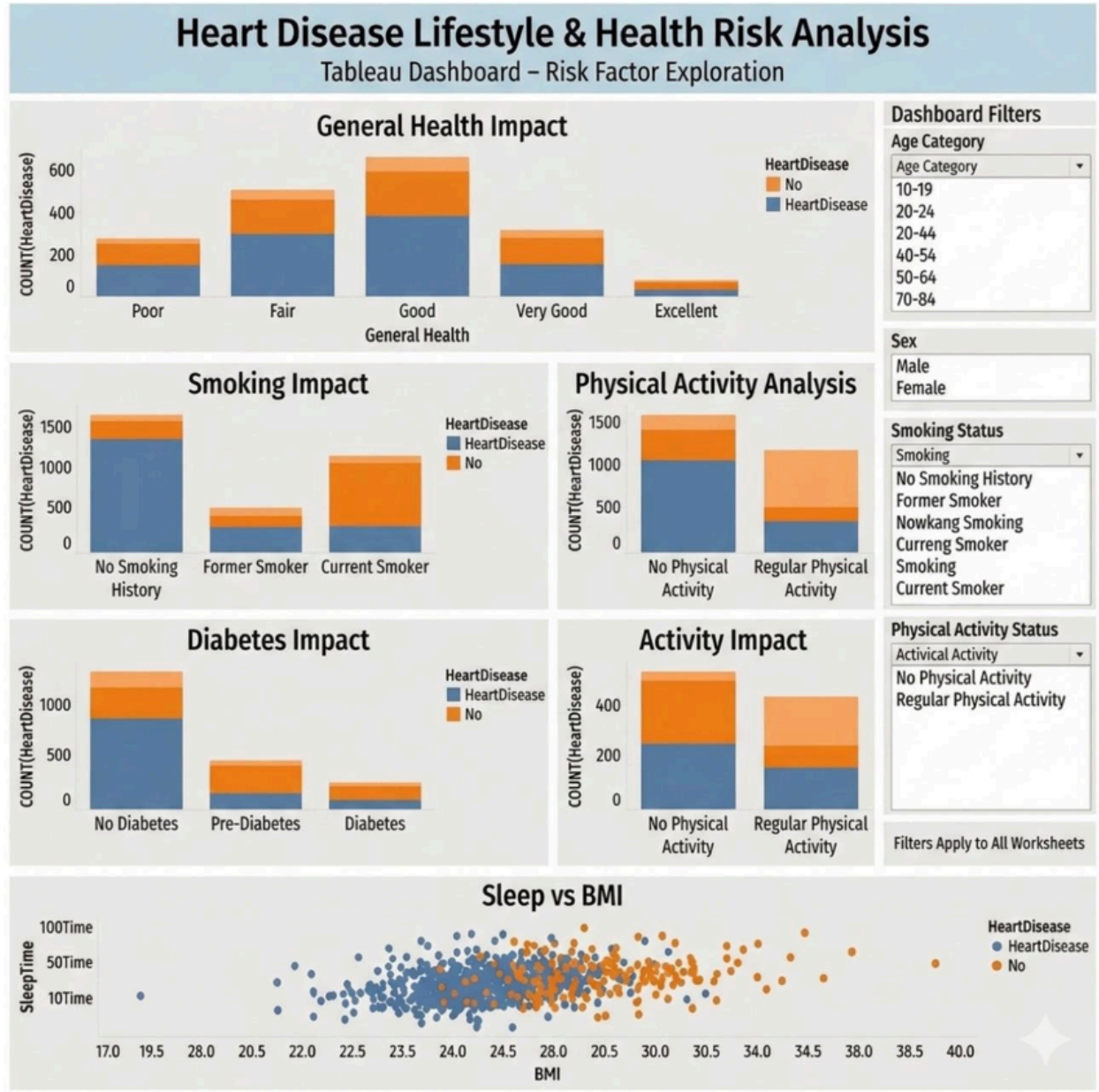
- Heart disease cases increase with age.
- Smoking is strongly associated with heart disease.
- Higher BMI values show a higher risk of heart disease.
- Physical activity appears to reduce heart disease risk.

The results obtained from the dashboard highlight the importance of lifestyle factors in influencing cardiovascular health. These insights can help healthcare researchers and professionals better understand the patterns associated with heart disease.

Overall, the results demonstrate that **data visualization using Tableau is an effective method for analyzing healthcare data and identifying important health trends**.

7.1 Output Screenshots:




Heart Disease Lifestyle & Health Risk Analysis
Tableau Dashboard – Risk Factor Exploration
General Health Impact
COUNT(HeartDisease)
600
400
200
0
Poor
Fair
Good
Very Good
Excellent
General Health
HeartDisease
No
HeartDisease
Smoking Impact
COUNT(HeartDisease)
1500
1000
500
0
No Smoking History
Former Smoker
Current Smoker
HeartDisease
HeartDisease
No
Physical Activity Analysis
COUNT(HeartDisease)
1500
1000
500
0
No Physical Activity
Regular Physical Activity
Diabetes Impact
COUNT(HeartDisease)
1000
500
0
No Diabetes
Pre-Diabetes
Diabetes
HeartDisease
HeartDisease
No
Activity Impact
COUNT(HeartDisease)
400
200
0
No Physical Activity
Regular Physical Activity
Dashboard Filters
Age Category
Age Category
10-19
20-24
20-44
40-54
50-64
70-84
Sex
Male
Female
Smoking Status
Smoking
No Smoking History
Former Smoker
Nowkang Smoking
Curreng Smoker
Smoking
Current Smoker
Physical Activity Status
Activical Activity
No Physical Activity
Regular Physical Activity
Filters Apply to All Worksheets
Sleep vs BMI
SleepTime
100Time
50Time
10Time
17.0
19.5
28.0
20.5
22.0
22.5
23.5
24.0
24.5
28.0
20.5
30.0
30.5
34.0
34.5
38.0
38.5
40.0
BMI
HeartDisease
HeartDisease
No

Examples:

Heart Disease by Age Category chart Smoking vs

Heart Disease chart

BMI vs Heart Disease visualization Final

Interactive Dashboard

8. ADVANTAGES & DISADVANTAGES

Advantages

Easy visualization of complex healthcare data.
Interactive dashboard for better exploration.
Helps identify major risk factors of heart disease.
Supports data-driven healthcare analysis.

The use of Tableau dashboards allows large and complex healthcare datasets to be presented in a clear and understandable format. Instead of analyzing raw data tables, users can easily observe patterns through visual charts and graphs.

Interactive dashboards also allow users to apply filters and explore the dataset from different perspectives. This makes the analysis process faster and more effective.

The project helps identify important risk factors such as **age, BMI, smoking habits, sleep duration, and physical activity**, which can contribute to heart disease. These insights can support healthcare professionals and researchers in understanding disease patterns.

Another advantage is that visual analytics improves decision-making by presenting information in an organized and easily interpretable way. It allows analysts to detect trends and correlations that may not be visible in raw data.

Disadvantages

Dataset may not represent all populations.
Limited to available health indicators.
Requires Tableau knowledge for development.

One limitation of the project is that the dataset used for analysis may not represent the entire global population. Therefore, the insights generated may not be applicable to every demographic group.

Another limitation is that the analysis is restricted to the health indicators available in the dataset. Important factors such as genetic history, diet patterns, or medical history may not be included, which could influence heart disease risk.

Additionally, the development and modification of dashboards require knowledge of Tableau software. Users who are not familiar with Tableau may need training to fully utilize the dashboard features.

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9. CONCLUSION

The Heart Disease Analysis project successfully demonstrates how data visualization can help understand complex healthcare data. By using Tableau dashboards, patterns and relationships between lifestyle habits and heart disease risk factors can be clearly observed.

The project highlights the importance of factors such as **smoking, BMI, age, sleep duration, and physical activity** in influencing heart disease risk. These factors play a significant role in determining cardiovascular health and can help identify individuals who may be at higher risk.

Through the use of interactive charts and filters, the Tableau dashboard allows users to explore the dataset and analyze different variables effectively. This interactive approach improves the understanding of healthcare data and supports efficient analysis.

Overall, the project shows that visualization tools can simplify complex datasets and provide meaningful insights that support healthcare research and preventive strategies. The results of this analysis can help healthcare professionals and researchers make informed decisions regarding heart disease prevention and awareness.

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10. FUTURE SCOPE

Future improvements for this project include:

Integrating machine learning models for heart disease prediction.
Expanding the dataset to include more health indicators.
Deploying dashboards on web platforms for wider accessibility.
Adding real-time healthcare data analysis.

In the future, machine learning algorithms can be integrated with the dataset to predict the likelihood of heart disease based on various health indicators. Predictive models could provide more advanced insights and assist healthcare professionals in early detection.

The dataset can also be expanded to include additional variables such as **diet habits, family medical history, stress levels, and cholesterol levels**, which could provide deeper insights into heart disease risk.

Another improvement could be deploying the Tableau dashboard on **web-based platforms or cloud services**, allowing a wider audience such as healthcare professionals, researchers, and policymakers to access the dashboard easily.

Real-time healthcare data integration could also be implemented in the future. This would allow the system to analyze continuously updated data and provide more accurate and timely insights about heart disease trends.

11. APPENDIX:

Dataset Link

https://drive.google.com/file/d/1hqJ6JDeWH6qtv2AligplkNfLd3Lq2FQe/view?usp=sharing

Source Code

Not applicable (Tableau-based visualization project).

GitHub & Project Demo Link

Github Link: https://github.com/Om-Prakash0/Heart_Disease_Analysis